# Phi-Based Quantum-Classical Routing: Empirical Evidence for a Crossover Regime on IBM Quantum Hardware

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**Abstract**

We investigate whether a training-free thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$, computed from IBM Quantum calibration data before circuit execution, can serve as a routing signal for allocating computation between quantum hardware and classical simulation in the tested setting. Across 23 tests on real IBM Quantum hardware, we validate a $\Phi$-based routing framework and show in direct crossover tests that LOW-$\Phi$ quantum execution can be worse than exact classical simulation for the tested classically tractable circuits: in the primary crossover test, classical simulation achieved 0.00% error while LOW-$\Phi$ quantum execution produced 8.23% error on the same circuit. Under a strict methodology with verified physical qubit binding and decisive ground truth, HIGH-$\Phi$ versus LOW-$\Phi$ error ratios ranged from 2.90× (QPE) to 30.47× (Bernstein-Vazirani) across 7 tested algorithms. The strict seven-algorithm tests constitute the primary evidence layer, while the broader 23-test suite provides supporting boundary conditions on size, depth, and reproducibility. Circuit size and depth impose additional routing boundaries: in the tested conditions, LOW-$\Phi$ execution at 15 qubits produced 94.85% error and execution at circuit depth 103 produced 67.29% error, both transitions into regimes where classical fallback was more reliable. These results support a routing decision framework in which minimum $\Phi$ across circuit qubits determines whether to execute on quantum hardware or fall back to classical simulation, with hybrid strategies retained as a natural extension.

## 1 Introduction

Quantum computing systems require decisions about when and where to execute circuits. On near-term quantum hardware, qubit quality varies significantly across devices and over time, and circuits assigned to poor-quality qubits can, in the tested setting, produce results that are not just degraded but worse than available classical simulation. Without a pre-execution quality check, quantum circuits are blindly assigned to hardware regardless of qubit state, wasting compute resources and producing unreliable outputs.

Prior work on qubit selection has focused on routing compiled circuits to lower-noise hardware subsets. Nation and Treinish (Nation and Treinish, 2023) developed mapomatic, which recovers approximately 40% of missing fidelity by routing circuits to low-noise hardware subgraphs after compilation. Their approach operates post-compilation and selects among available hardware assignments. The present work addresses a different decision: not which qubits to use within quantum hardware, but whether to use quantum hardware at all for a given circuit, based on pre-execution qubit quality measured by $\Phi$.

The stability metric at the center of this work is:

$$\Phi = I \times \rho - \alpha \times S \tag{1}$$

where $I = (F - 0.50)/0.50$ is normalized gate fidelity above the random baseline for two-level systems, $\rho = \min(T_2/T_1, 1.0)$ is the coherence ratio, $S$ is readout error, and $\alpha = 0.1$ is a coupling constant. The threshold $\Phi_c = 0.25$ provides a baseline GOOD ($\Phi \geq 0.25$) or NOT-GOOD ($\Phi < 0.25$) partition for routing. Negative $\Phi$ indicates a collapse or failure regime. Prior work validated this formula across 445 qubits on three IBM backends, establishing that $\Phi$ tracks coherence ratio $\rho = T_2/T_1$ at $r = 0.9458$ and that $\Phi$-based qubit selection reduces circuit error by 83% in the tested setting (Barnicle, 2026a).

This paper addresses a different question: can $\Phi$ serve as a routing criterion that determines not just which qubits to use but whether to use quantum hardware at all? The central finding is that there is a real crossover regime in the tested setting where LOW-$\Phi$ quantum execution can be worse than exact classical simulation. The crossover claim is bounded to classically tractable circuits in the tested IBM setting and should not be read as a claim that classical simulation is preferable for all quantum workloads. The minimum $\Phi$ across circuit qubits is the operative routing quantity, and circuit size and depth are additional routing variables that can drive execution into noise-dominated regimes regardless of individual qubit quality.

The main contributions of this paper are:

1. Empirical evidence for a quantum-classical crossover regime in the tested classically tractable circuits, where LOW-$\Phi$ quantum execution produces higher error than exact classical simulation, together with broader routing evidence from a test suite spanning 7 algorithms and 3 IBM backends.

2. A weakest-link routing principle: circuit quality is governed by the minimum $\Phi$ across participating qubits, and the routing decision should be based on that minimum rather than average or maximum qubit quality.

3. Characterization of circuit size and depth as additional routing variables, with documented transition into noise-dominated and worse-than-random error regimes in the tested LOW-$\Phi$ conditions, including 94.85% error at 15 qubits and 67.29% error at depth 103.

4. A strict validation methodology for routing claims, requiring verified physical qubit binding, decisive ground truth, and true LOW-$\Phi$ qubit selection, which materially strengthened the evidence relative to the earlier research-grade tests.

# 2 Related Work

Quantum hardware quality assessment and qubit selection have been studied primarily through calibration metrics and post-compilation routing. Nation and Treinish (Nation and Treinish, 2023) developed mapomatic, a post-compilation layout optimizer that scores qubit subsets by multiplying individual error rates. Their work demonstrated approximately 40% fidelity recovery through hardware-aware routing. Their cost function explicitly excluded $T_1/T_2$-derived coherence information because it did not materially affect layout ordering in their single-snapshot experiments. The present work studies a different routing context. Prior work using the same $\Phi$ formulation found that the coherence ratio $\rho = T_2/T_1$ is the dominant component in cross-backend stability classification, which is consistent with the crossover behavior observed here in the hybrid routing setting. The present paper asks whether the resulting $\Phi$ score can be used as a routing signal for execution allocation.

Quantum error mitigation and noise characterization have received substantial attention. Temme et al. (Temme et al., 2017) introduced zero-noise extrapolation as a method for mitigating coherent and incoherent errors by scaling noise and extrapolating to the zero-noise limit. These methods operate during or after execution and do not address the pre-execution routing question of whether to use quantum hardware at all.

Transfer learning and domain generalization methods provide a framework for thinking about cross-backend stability. Pan and Yang (Pan and Yang, 2010) survey transfer learning across domains. Prior work using the same $\Phi$ formulation reported that $\Phi$-component-based qubit quality prediction transfers across IBM backends under backend-split evaluation, supporting the use of $\Phi$-related signals as cross-backend routing features (Barnicle, 2026b).

Prior work on hybrid quantum-classical computation has largely focused on algorithmic hybridization: variational quantum algorithms (Cerezo et al., 2021) decompose problems so that parameterized quantum circuits are optimized by a classical outer loop. The present work addresses a different kind of hybrid allocation: a pre-execution routing decision that sends entire circuits to quantum or classical execution based on hardware quality, rather than decomposing the algorithm itself.

Classical simulation remains tractable for the small tested circuits used in the crossover experiments and for a broader range of small to moderate system sizes discussed in prior work (Villalonga et al., 2019). The crossover regime identified in this paper is relevant precisely where classical simulation remains tractable but quantum execution is unreliable due to poor qubit quality.

# 3 Methods

## 3.1 The Phi Stability Metric

The routing framework uses the thermodynamic stability metric:

$$\Phi = I \times \rho - \alpha \times S \tag{2}$$

where the components are computed from IBM Quantum calibration data as follows. Normalized fidelity is $I = (F - 0.50)/0.50$, where $F = 1 - \text{single-qubit gate error}$ is single-qubit gate fidelity and 0.50 is the random baseline for two-level systems. Coherence ratio is $\rho = \min(T_2/T_1, 1.0)$, where $T_2$ is coherence time and $T_1$ is relaxation time. Readout error is $S = E$, the calibration-reported readout error rate. The coupling constant is $\alpha = 0.1$ and the routing threshold is $\Phi_c = 0.25$. All circuits and transpilation steps were implemented in Qiskit, hardware execution used IBM Quantum Runtime, and classical simulation used Qiskit's exact statevector simulator.

For the routing decision, the operative quantity is the minimum $\Phi$ across all qubits participating in the circuit:

$$\Phi_{\text{min}} = \min_{i \in \text{circuit qubits}} \Phi_i \tag{3}$$

The routing decision function is:

$$\text{route} = \begin{cases} \text{QUANTUM} & \text{if } \Phi_{\text{min}} \geq 0.25 \\ \text{CLASSICAL} & \text{if } \Phi_{\text{min}} < 0.25 \text{ and circuit is classically tractable} \\ \text{HYBRID} & \text{otherwise} \end{cases} \tag{4}$$

This paper validates the QUANTUM versus CLASSICAL routing boundary in the tested setting. The HYBRID branch is included as part of the broader allocation framework but is not directly evaluated in the experiments reported here.

## 3.2 IBM Quantum Hardware and Calibration Data

Calibration data and circuit execution used three IBM Quantum backends: ibm_fez (156 qubits), ibm_torino (133 qubits), and ibm_marrakesh (156 qubits). For each qubit, the calibration record contains single-qubit gate fidelity $F$, relaxation time $T_1$, coherence time $T_2$, and readout error $E$. Phi was computed from these calibration fields for each qubit before each experiment. No synthetic data were used at any stage.

## 3.3 Research-Grade and Strict Test Methodology

The 23 validation tests are divided into two layers. The strict algorithm tests provide the primary routing evidence, while the size-scaling and depth-scaling tests provide supporting boundary-condition evidence.

The 16 research-grade tests (Tests 1-16) used standard qubit selection and basic ground truth checks. LOW-$\Phi$ qubit selection in the research tests used adjacent-ID fallback when connected low-quality qubits were not available. These tests provided initial routing evidence and boundary-condition results on circuit size and depth.

The 7 strict tests (Tests 17-23) introduced four additional controls that substantially strengthened the evidence. First, physical qubit binding was verified via `q.index` after transpilation, confirming that the transpiler did not remap the intended qubits. Tests were aborted if remapping occurred. Second, ground truth circuits were required to achieve at least 95% decisive fidelity in preflight validation or the test was aborted. Third, LOW-$\Phi$ qubits were selected from connected qubit pools only, with negative-$\Phi$ qubits preferred when available. Fourth, a two-mode preflight-then-execute protocol was used: preflight validation confirmed qubit binding, ground truth, and layout before hardware execution was submitted.

The strict tests are the primary evidence layer for this paper. The research tests provide supporting boundary-condition evidence on size, depth, and multi-backend behavior.

## 3.4 Algorithms Tested

Seven quantum algorithms were tested under the strict methodology: Quantum Phase Estimation (QPE), Grover's Algorithm, the Deutsch-Jozsa Algorithm, the Bernstein-Vazirani Algorithm, Simon's Algorithm, the Quantum Fourier Transform (QFT), and GHZ state preparation. These algorithms were selected to cover structurally distinct circuit families: phase-estimation-based circuits (QPE), oracle-based circuits (Grover, Deutsch-Jozsa, Bernstein-Vazirani, Simon's), transform-based circuits (QFT), and entanglement circuits (GHZ).

For each algorithm test, HIGH-$\Phi$ qubits were selected from the available pool by choosing qubits with $\Phi_{\min} \geq 0.25$, with preference for the highest available $\Phi$ values. LOW-$\Phi$ qubits were selected with preference for negative-$\Phi$ values when available. Simon's Algorithm (Test 21) required a fallback LOW selection because no negative-$\Phi$ connected quad was available on the tested backends at the time of the experiment.

### 3.5 Error Measurement and Ground Truth

For all circuit executions, error was measured as deviation from the expected correct output. For binary-output circuits such as Bernstein-Vazirani and Deutsch-Jozsa, error was the fraction of shots that did not return the correct bitstring. For GHZ and QFT circuits, error was measured as 1 minus the fidelity of the dominant output state relative to the expected correct state. For the classically tractable circuits used in the crossover tests, classical reference results were obtained from noiseless statevector simulation, which yields zero execution error by construction for deterministic circuits; task-specific comparison metrics are reported in the individual test descriptions where applicable.

Research tests typically used 8,192 shots and strict tests typically used 4,096 shots, unless otherwise noted in the individual test descriptions.

### 3.6 Size and Depth Scaling Tests

To characterize routing boundaries beyond the algorithm-specific tests, dedicated size-scaling and depth-scaling experiments were conducted across the tested qubit-quality conditions. Size scaling used GHZ-style circuits at 3, 5, 7, 10, 15, and 20 qubits to measure how error changes with qubit count across the tested qubit selections. Depth scaling used identity-return circuits at depths 3, 13, 23, 53, 103, and 203 gates to measure how error changes with circuit depth in the tested setting.

These tests used research-grade methodology and are reported as boundary-condition evidence rather than primary routing validation.

## 4 Results

### 4.1 Quantum-Classical Crossover

In the primary crossover test, a 3-qubit GHZ circuit was executed on HIGH-$\Phi$ qubits (ibm_fez, min-$\Phi$ = 0.999) and LOW-$\Phi$ qubits (ibm_marrakesh, min-$\Phi$ = 0.052), with classical simulation providing the reference baseline.

Table 1: Primary Crossover Test: 3-Qubit GHZ Circuit

| Execution Mode | min-$\Phi$ | Error (%) |
|---|---|---|
| Classical simulation | N/A | 0.00 |
| HIGH-$\Phi$ quantum | 0.999 | 2.87 |
| LOW-$\Phi$ quantum | 0.052 | 8.23 |

LOW-Φ quantum execution produced 8.23% error compared to 0.00% for classical simulation on the same circuit, confirming the crossover in the tested setting. HIGH-Φ quantum execution remained low-error at 2.87%. The same crossover pattern was also observed in additional tested cases, including QFT (classical 2.10% versus LOW-Φ 16.65%), Grover (classical 0.00% versus LOW-Φ 13.75%), Deutsch-Jozsa (classical 0.00% versus LOW-Φ 7.54%), and strict Bernstein-Vazirani (classical 0.00% versus LOW-Φ 64.72%).

## 4.2 Strict Algorithm Results

Under the strict methodology with verified physical qubit binding and negative-Φ LOW qubit selection where available, all seven tested algorithms showed clear HIGH-Φ versus LOW-Φ error separation. Where paired research-grade comparisons exist, the strict methodology produced materially stronger separations.

Table 2: Strict Test Results Across Seven Algorithms (Tests 17-23)

| Algorithm | Test | HIGH-Φ Error (%) | LOW-Φ Error (%) | Ratio |
|---|---|---|---|---|
| Bernstein-Vazirani | 20 | 2.12 | 64.72 | 30.47× |
| Grover | 18 | 5.76 | 91.60 | 15.90× |
| Deutsch-Jozsa | 19 | 5.76 | 74.73 | 12.97× |
| QFT | 22 | 1.66 | 13.48 | 8.12× |
| GHZ | 23 | 3.88 | 28.66 | 7.38× |
| Simon's | 21 | 1.76 | 7.15 | 4.07× |
| QPE | 17 | 5.76 | 16.70 | 2.90× |

Ratios ranged from 2.90× (QPE) to 30.47× (Bernstein-Vazirani). All seven algorithms showed the same direction: HIGH-Φ qubits produced substantially lower error than LOW-Φ qubits. The method is therefore algorithm-agnostic in direction across the seven tested circuit families in the tested setting, though not uniform in effect size.

The strict methodology materially strengthened the evidence relative to the research-grade tests on the same algorithms. Grover improved from 3.27× under research methodology to 15.90× under strict methodology. This improvement resulted from the strict tests using truly negative-Φ LOW selections when available rather than merely below-threshold qubits. QPE adds a seventh strict-only validation with no earlier research-grade counterpart.

Simon's Algorithm (Test 21) is the weakest strict result at 4.07× because it required a fallback LOW selection: no negative-Φ connected quad was available on the tested backends at the time of the experiment, so a connected below-threshold selection was used instead.

## 4.3 Weakest-Link Routing Principle

Tests 2 and 3 provide the clearest internal evidence for the weakest-link principle. Both tests used identical GHZ circuits and the same backend, varying only the quality of the selected qubits within the above-threshold region.

Table 3: Weakest-Link Evidence: First-Above-Threshold Versus Best-Available Qubit Selection

| Selection Strategy | min-$\Phi$ | HIGH-vs-LOW Error Ratio |
|---|---|---|
| First available above threshold | 0.254 | $\approx 1\times$ |
| Best available above threshold | 0.999 | $2.66\times$ |

Using the first qubits that cleared the 0.25 threshold at min-$\Phi = 0.254$ produced only marginal improvement over LOW-$\Phi$ execution. Using the best available HIGH-$\Phi$ qubits at min-$\Phi = 0.999$ produced a clear $2.66\times$ advantage. This result shows that the routing decision should depend on the minimum $\Phi$ across circuit qubits and on how far that minimum lies above the threshold, not merely on whether the threshold is cleared. Merely meeting the threshold criterion is not sufficient for reliable quantum execution.

## 4.4 Threshold Sweep

Test 6 measured error across five representative points in the $\Phi$ spectrum to characterize the relationship between $\Phi$ value and execution quality.

Table 4: Threshold Sweep: Error Across the $\Phi$ Spectrum (ibm_fez, GHZ Circuit)

| min-$\Phi$ | Error (%) |
|---|---|
| −0.050 | 55.32 |
| 0.210 | 60.33 |
| 0.474 | 7.64 |
| 0.730 | 39.97 |
| 0.999 | 3.17 |

Error varied approximately $19\times$ across the tested $\Phi$ spectrum, from 3.17% at $\Phi = 0.999$ to 60.33% at $\Phi = 0.210$. The relationship is not monotonic across all tested values: $\Phi = 0.474$ produced 7.64% error while $\Phi = 0.730$ produced 39.97%, a reversal consistent with backend-specific gate calibration factors that can affect execution quality independently of single-qubit $\Phi$. The most important negative finding in this test is the above-threshold case at $\Phi = 0.730$, which still produced 39.97% error. This establishes that the 0.25 threshold is a necessary but not sufficient condition for reliable execution. The threshold is a useful routing boundary, not a guarantee of low error.

## 4.5 Circuit Size Scaling

Tests 5, 7, 10, and 14 measured how error changes with qubit count across the tested qubit-quality regimes.

Table 5: Circuit Size Scaling Across the Tested Qubit-Quality Conditions

| Qubits | min-$\Phi$ | Error (%) | Status |
|---|---|---|---|
| 3 | 0.999 | $\approx$3 | Usable |
| 5 | 0.997 | 7.20 | Marginal |
| 7 | 0.774 | 48.78 | Near-random |
| 10 | 0.672 | 22.29 | High error |
| 15 | 0.569 | 94.85 | Noise-dominated |
| 20 | 0.436 | 92.50 | Noise-dominated |

Execution quality deteriorated substantially as circuit size increased in the tested conditions, with severe failure appearing at 15 and 20 qubits. The trend was not strictly monotonic at every intermediate size: the 10-qubit result at 22.29% is lower than the 7-qubit result at 48.78%, an anomaly likely attributable to the specific qubits available at the time of the test rather than a fundamental reversal in the size scaling relationship. At 15 and 20 qubits, errors of 94.85% and 92.50% indicate noise-dominated execution in the tested conditions. These results support circuit size as an additional routing variable beyond $\Phi$ alone.

## 4.6 Circuit Depth Scaling

Test 8 and related depth-scaling experiments measured how error grows with circuit depth using identity-return circuits. Shallow depth results (3, 13, 23 gates) and extreme-depth results (53, 103, 203 gates) are reported together here to show the overall trend, but they were not all collected under one identical qubit-selection condition.

Table 6: Circuit Depth Scaling in the Tested Setting

| Depth (gates) | Error (%) | Status |
|---|---|---|
| 3 | $\approx$3 | Usable |
| 13 | $\approx$5 | Marginal |
| 23 | $\approx$8 | Marginal |
| 53 | 22.00 | High error |
| 103 | 67.29 | Worse than random |
| 203 | 64.06 | Worse than random |

At depth 103, error reached 67.29%, which exceeds 50% and is therefore worse than random guessing for binary-style measurement tasks in the tested setting. At depth 203, error was 64.06%. For these tested classically tractable circuits in this regime, classical simulation was more reliable than the observed quantum execution results. These results support circuit depth as an additional routing variable in the tested setting.

## 4.7 Reproducibility and Multi-Backend Support

Test 15 repeated the primary crossover test (Test 1) using the same qubit selection procedure to assess result stability.

Table 7: Reproducibility Test: Primary Crossover Repeated

| Condition | Run 1 Error (%) | Run 2 Error (%) |
|---|---|---|
| HIGH-Φ quantum | 2.87 | 2.37 |
| LOW-Φ quantum | 8.23 | 8.11 |

The same qubits were selected in both runs. HIGH-Φ error changed by 0.50 percentage points and LOW-Φ error changed by 0.12 percentage points across the two runs, confirming that the crossover effect is reproducible in the primary tested case and is not a one-off outcome.

The routing framework was also validated across all three tested IBM backends: ibm_fez, ibm_torino, and ibm_marrakesh. Test 4 confirmed that the HIGH-Φ versus LOW-Φ routing direction held consistently across all three backends, with backend-specific error magnitudes varying but the routing direction remaining the same.

# 5 Discussion

The central finding of this paper is that a training-free thermodynamic stability metric Φ, computed from IBM Quantum calibration data before circuit execution, can identify a crossover regime in the tested setting where LOW-Φ quantum execution is worse than exact classical simulation. This is not just a discrimination result: it is a routing result. In the tested classically tractable setting, one appropriate engineering response when $\Phi_{\min}$ falls below the 0.25 threshold is to route the computation to classical simulation rather than proceed directly with quantum execution.

The strict methodology introduced in this work materially strengthened the evidence. Requiring verified physical qubit binding, decisive ground truth, and true negative-Φ LOW qubit selection raised the Grover separation from 3.27× under research conditions to 15.90× under strict conditions. The 30.47× Bernstein-Vazirani result, the strongest in the suite, was achievable only because the strict protocol ensured that the LOW-Φ qubits were genuinely low-quality rather than merely below a soft threshold. This finding has methodological implications beyond this paper: credible hardware routing claims of this kind benefit from strict qubit-binding verification rather than relying only on requested layouts or soft threshold checks.

The weakest-link principle, that circuit quality is governed by the minimum Φ across participating qubits, is supported by the Test 2 versus Test 3 comparison and by the size-scaling results. Qubits that barely clear the 0.25 threshold at min-Φ = 0.254 still produce approximately 1× error ratio, while qubits at min-Φ = 0.999 produce 2.66×. This means the routing decision should depend on the minimum Φ across circuit qubits and on how far that minimum lies above the threshold, not merely on whether the threshold is cleared. A qubit pool with many marginal qubits may still produce unreliable circuit execution even when the routing rule nominally allows quantum execution.

The threshold sweep result at Φ = 0.730 with 39.97% error is the most practically important negative finding in the paper. It establishes that the 0.25 threshold is a necessary but not sufficient condition for reliable execution. Backend-specific factors including gate calibration quality independent of single-qubit Φ can produce elevated error even for above-threshold qubits. The routing framework therefore works best in combination with best-available qubit selection rather than as a hard binary threshold applied without regard to the quality margin.

The size and depth scaling results establish two additional routing variables beyond $\Phi$ alone. At 15 and 20 qubits in the tested conditions, error reached 94.85% and 92.50%, indicating noise-dominated execution where quantum output carries little reliable information in those tested conditions. At depth 103, error reached 67.29%, exceeding the worse-than-random boundary for binary measurement tasks. These results suggest that the routing decision function should incorporate circuit size and depth as inputs alongside $\Phi_{\min}$, not treat qubit quality as the sole routing criterion.

The comparison to mapomatic (Nation and Treinish, 2023) highlights the difference in approach. Mapomatic operates post-compilation and selects the best available hardware assignment among quantum options. The present work operates pre-execution and answers a prior question: is quantum execution the right choice at all for this circuit, given the current hardware state? These two approaches are complementary rather than competing. A complete pre-execution routing system could apply $\Phi$-based quantum versus classical routing first, and then apply post-compilation qubit selection optimization within the quantum branch.

# 6 Limitations

The results in this paper are bounded to three IBM Quantum backends (ibm_fez, ibm_torino, ibm_marrakesh) and to the specific circuits and qubit selections tested in January 2026. Whether the crossover regime, the weakest-link principle, and the size and depth boundaries generalize to other IBM backends, to other quantum hardware platforms (Google, IonQ, Rigetti), or to circuits collected under different hardware conditions is not established by this study.

The crossover claim is bounded to classically tractable circuits where exact or near-exact classical simulation is available. The routing framework does not claim that classical simulation is preferable for all quantum workloads. For circuits that are classically intractable, the HYBRID branch of the routing framework is the appropriate fallback, but the HYBRID branch was not directly evaluated in the experiments reported here.

The threshold sweep result at $\Phi = 0.730$ with 39.97% error shows that the 0.25 threshold alone is not sufficient to guarantee low-error execution. Backend-specific gate calibration quality independent of single-qubit $\Phi$ can still produce elevated error even for above-threshold qubits. The threshold is a useful baseline routing boundary, not a calibrated guarantee.

The size-scaling trend is not strictly monotonic: the 10-qubit result at 22.29% error is lower than the 7-qubit result at 48.78%. This non-monotonicity likely reflects variation in the specific qubits available at the time of each test rather than a fundamental reversal in the scaling relationship, but it means the size scaling should be interpreted as a general trend in the tested conditions rather than a calibrated boundary function.

The depth-scaling results at different depths were not all collected under one identical qubit-selection condition. The reported trend from usable at shallow depth to worse-than-random at depth 103 is supported by the evidence, but the exact depth at which the transition occurs will depend on hardware, qubit quality, and circuit structure.

Simon's Algorithm (Test 21) required a fallback LOW selection because no negative-$\Phi$ connected quad was available on the tested backends at the time of the experiment. Its

4.07× ratio is the weakest of the strict results and should be interpreted as the lower bound of the strict suite rather than the central estimate.

The reproducibility result covers only the primary GHZ crossover test. Reproducibility was not evaluated exhaustively across all 23 tests.

# 7 Conclusion

We investigated whether the training-free thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$, computed from IBM Quantum calibration data before circuit execution, can serve as a routing signal for allocating computation between quantum hardware and classical simulation. Across 23 tests spanning 7 algorithms and 3 IBM backends, we find:

1. In the tested classically tractable circuits, LOW-Φ quantum execution can be worse than exact classical simulation. The primary crossover test showed 8.23% LOW-Φ error versus 0.00% classical error on the same GHZ circuit.

2. Under a strict methodology with verified physical qubit binding and negative-Φ LOW qubit selection, HIGH-Φ versus LOW-Φ error ratios ranged from 2.90× (QPE) to 30.47× (Bernstein-Vazirani) across 7 tested algorithms, supporting algorithm-agnostic routing evidence in the tested setting.

3. Circuit quality is governed by the minimum Φ across participating qubits. Qubits barely above the 0.25 threshold produce approximately 1× error ratio while best-available qubits produce 2.66×, showing the routing decision should depend on the quality margin above the threshold.

4. Circuit size and depth impose additional routing boundaries in the tested conditions. At 15 qubits, error reached 94.85%; at depth 103, error reached 67.29%, both in noise-dominated or worse-than-random regimes.

5. The strict validation methodology produced substantially stronger evidence than the earlier research-grade tests, showing that verified qubit binding materially strengthens routing evidence relative to soft threshold checks.

These results support a routing decision framework in which minimum Φ across circuit qubits, together with circuit size and depth, determines whether to execute on quantum hardware or fall back to classical simulation for classically tractable circuits. The threshold of 0.25 provides a useful baseline routing boundary but not a guarantee of low error: backend- specific factors can produce elevated error even for above-threshold qubits, and the routing decision is most reliable when combined with best-available HIGH-Φ qubit selection.

# Code Availability

Code and data are available at: `https://github.com/Wise314/phi-hybrid-allocation`.

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IBM Quantum hardware accessed via IBM Quantum Network.

## Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/956,752 (filed January 9, 2026). No license to implement or commercialize the described methods is granted by this publication. 

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